

Tuna Kimyonok

Backend & Applied AI Engineer

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SUMMARY

I build end-to-end backend and applied AI systems and optimize them for speed and cost. During my last internship I built a voice assistant that answers from Turkish national curriculum (MEB) textbooks with its sources shown, from the ESP32-S3 firmware to the FastAPI backend, and cut time to first audio from 11 seconds to 3. I graduate in June 2027 and am available to work full-time.

Remote, hybrid or on-site, and able to relocate within Türkiye or abroad.

EXPERIENCE

Software Engineering Intern · THINKBRO

Jul 2026 – Sep 2026

- Built a voice AI assistant that answers from Turkish national curriculum (MEB) textbooks end to end: the ESP32-S3 firmware, the FastAPI backend, and an 8,577-chunk Qdrant knowledge base covering all 12 grade levels.
- Cut time-to-first-audio from 11.0 s to 2.7–3.8 s by making the whole chain streaming (STT → hybrid search → LLM → TTS) and sending each sentence to be spoken as soon as its boundary was reached.
- Blocked hallucination with a three-layer defence — a prompt rule, a deterministic KAYNAK_YOK marker and two-way automated evaluation — measuring 6/6 correct refusals on out-of-scope questions and 6/6 correct answers on in-scope ones.
- Brought up the ESP32-S3 peripherals: the ILI9341 touch display over SPI, and the INMP441 microphone and MAX98357A speaker over I2S.
- Chose a file-based transcription API over a streaming one after realising push-to-talk delivers the audio in one piece, lowering cost without giving up the latency target.

PROJECTS

Steam Player Feedback Analysis System

Mar 2026 – May 2026

Python · Flask · Sentence Transformers · UMAP · HDBSCAN · ChromaDB · Gemini 2.5 Flash

- Engineered an NLP data pipeline to ingest and process 15,000+ unstructured user reviews, using UMAP/HDBSCAN clustering to automatically categorize technical issue reports.
- Integrated a ChromaDB vector database with a Gemini 2.5 Flash RAG framework to support natural language querying and dynamic metadata filtering.
- Enhanced backend pipeline resilience by implementing automated API rate-limiting and key-rotation mechanisms, plus a scoring algorithm for issue prioritization.

Dwarf Grom: AI-Driven NPC Prototype

Oct 2025 – Dec 2025

Unity 6 · C# · Python · FastAPI · Gemini 2.5 Flash · ElevenLabs · Google Speech Recognition

- Architected an asynchronous RESTful API connection between a Python (FastAPI) backend and a Unity 6 client, handling structured JSON payloads and Base64 audio streaming at low latency.
- Implemented backend evaluation logic powered by Gemini 2.5 Flash that scores the player's own words and drives game-state transitions accordingly.

SKILLS

Languages	Python · C# · JavaScript · C / C++
Backend & APIs	FastAPI · Flask · REST API design · WebSocket · Server-Sent Events · asyncio
AI & Retrieval	RAG architecture · Qdrant (hybrid search) · ChromaDB · Sentence Transformers · Gemini API · OpenAI API · UMAP + HDBSCAN · Structured LLM output (JSON)
Data & Tooling	PostgreSQL · Docker Compose · Git & GitHub · Postman
Game & Embedded	Unity 6 · ESP32-S3 (ESP-IDF)
Languages	Turkish (Native) · English (Professional working proficiency)

EDUCATION

B.S. in Computer Engineering · Iskenderun Technical University

Sep 2022 – Jun 2027 (expected)